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#include <WiFi.h>
#include <WebSocketsServer.h>
#include <HTTPClient.h>
#include <WiFiClientSecure.h>

// ✓ YOUR WIFI
const char* ssid = "iot5";
const char* password = "123456789";

// ☒ TELEGRAM
String BOT1 = "8704029155:AAFokocwJdR-a2lo2rtc__RFZSse_KldUQY"; // ☒ Main pipeline bot
String BOT2 = "7650350789:AAEp3NUgRthfuYJd7AzMDnSDz97iZp-YpxY"; // ☒ Sub pipeline bot
String CHAT_ID = "6369522076";
// WebSocket
WebSocketsServer webSocket = WebSocketsServer(81);

// Pins
#define F1 34
#define F2 35
#define F3 32
#define F4 33
#define TRIG 5
#define ECHO 17
#define RELAY 26

// Flow counters
volatile int p1=0,p2=0,p3=0,p4=0;

// Values
float f1,f2,f3,f4;
int tank;

int tankHeight = 100;

// Status memory (avoid spam)
String lastMainStatus="";
String lastSubStatus="";

// Interrupts
void IRAM_ATTR isr1(){p1++;}
void IRAM_ATTR isr2(){p2++;}
void IRAM_ATTR isr3(){p3++;}
void IRAM_ATTR isr4(){p4++;}

// Tank
int readTank(){
    digitalWrite(TRIG, LOW); delayMicroseconds(2);
    digitalWrite(TRIG, HIGH); delayMicroseconds(10);
    digitalWrite(TRIG, LOW);

    long d = pulseIn(ECHO, HIGH, 30000);
    float dist = d * 0.034 / 2;

    int per = ((tankHeight - dist)/tankHeight)*100;

    if(per < 0) per = 0;
    if(per > 100) per = 100;

    return per;
}

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// Telegram send
void sendTelegram(String bot, String msg){

    WiFiClientSecure client;
    client.setInsecure();

    HTTPClient https;

    String url = "https://api.telegram.org/bot" + bot +
        "/sendMessage?chat_id=" + CHAT_ID +
        "&text=" + msg;

    https.begin(client, url);
    https.GET();
    https.end();

    Serial.println("📡 " + msg);
}

// WebSocket
void websocketEvent(uint8_t num, WStype_t type, uint8_t * payload, size_t length){
    if(type == WStype_CONNECTED) Serial.println("✔️ Client Connected");
    if(type == WStype_DISCONNECTED) Serial.println("❌ Client Disconnected");
}

// ===== SETUP =====
void setup(){

    Serial.begin(115200);
    delay(1500);

    Serial.println("\n🔧 Starting System...");

    pinMode(F1, INPUT);
    pinMode(F2, INPUT);
    pinMode(F3, INPUT);
    pinMode(F4, INPUT);

    pinMode(TRIG, OUTPUT);
    pinMode(ECHO, INPUT);

    pinMode(RELAY, OUTPUT);
    digitalWrite(RELAY, HIGH); // Motor OFF

    attachInterrupt(F1, isr1, RISING);
    attachInterrupt(F2, isr2, RISING);
    attachInterrupt(F3, isr3, RISING);
    attachInterrupt(F4, isr4, RISING);

    // 🔄 CONNECT WIFI FIRST
    WiFi.begin(ssid, password);

    Serial.print("Connecting WiFi");

    while(WiFi.status() != WL_CONNECTED){
        delay(1000);
        Serial.print(".");
    }
}

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Serial.println("\n✔WiFi Connected");
Serial.print("IP: ");
Serial.println(WiFi.localIP());

WebSocket.begin();
WebSocket.onEvent(WebSocketEvent);

Serial.println("✔ WebSocket Started");
}

// ===== LOOP =====
void loop(){

    WebSocket.loop();

    // Flow
    f1 = p1 * 0.00225;
    f2 = p2 * 0.00225;
    f3 = p3 * 0.00225;
    f4 = p4 * 0.00225;

    p1=p2=p3=p4=0;

    tank = readTank();

    float diffM = abs(f1 - f2);
    float diffS = abs(f3 - f4);

    String m_status, s_status;

    // MAIN
    if(diffM < 0.02) m_status="NORMAL";
    else if(diffM < 0.035) m_status="THEFT";
    else m_status="DAMAGE";

    // SUB
    if(diffS < 0.02) s_status="NORMAL";
    else if(diffS < 0.08) s_status="LEAK";
    else s_status="FULL_DAMAGE";

    //  RELAY MOTOR CONTROL
    String motor;

    if(tank < 25 || m_status=="DAMAGE" || s_status=="FULL_DAMAGE"){
        motor = "OFF";
        digitalWrite(RELAY, HIGH);
    }else{
        motor = "ON";
        digitalWrite(RELAY, LOW);
    }

    //  TELEGRAM ALERTS

    if(m_status != lastMainStatus){
        if(m_status=="THEFT")
            sendTelegram(BOT1,"⚠ MAIN PIPE THEFT");
        else if(m_status=="DAMAGE")
            sendTelegram(BOT1,"⚠ MAIN PIPE DAMAGE");
    }
}

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        lastMainStatus = m_status;
    }

    if(s_status != lastSubStatus){
        if(s_status=="LEAK")
            sendTelegram(BOT2,"⚠ SUB PIPE LEAK");
        else if(s_status=="FULL_DAMAGE")
            sendTelegram(BOT2,"💣 SUB PIPE FULL DAMAGE");

        lastSubStatus = s_status;
    }

    // SERIAL
    Serial.println("\n--- DATA ---");
    Serial.print("Tank: "); Serial.println(tank);
    Serial.print("Main: "); Serial.println(m_status);
    Serial.print("Sub: "); Serial.println(s_status);
    Serial.print("Motor: "); Serial.println(motor);

    // JSON
    String json="{";
    json+="\"tank\": "+String(tank)+",";
    json+="\"motor\": \""+motor+"\", ";
    json+="\"m_in\": "+String(f1)+",";
    json+="\"m_out\": "+String(f2)+",";
    json+="\"m_status\": \""+m_status+"\", ";
    json+="\"s_in\": "+String(f3)+",";
    json+="\"s_out\": "+String(f4)+",";
    json+="\"s_status\": \""+s_status+"\"";
    json+="}";

    websocket.broadcastTXT(json);

    delay(2000);
}

```